

Yang Zhongyu

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RESEARCH INTERESTS

Fields: Multimodal Learning, MultiAgent, Vision-Language Models, Generative AI

Topics: MLLM; MultiAgent; Diffusion-based Generation; Retrieval-Augmented Reasoning

Objective: My long-term goal is to build general-purpose multimodal systems that can perceive, reason, and communicate effectively across visual, textual, and behavioral modalities in dynamic, real-world environments.

EDUCATION

- **Lanzhou University (Project 985)** Sept. 2021 - June.2025
B.S. in Mathematics(the Basic Theory Class)(Main major) and Management (Minor) Lanzhou, China
 - Relevant courses: Mathematical Analysis, Advanced Algebra, Python, C++, Probability Theory, Ordinary Differential Equations, Numerical Analysis, Microeconomics, Differential Geometry, Functional Analysis, etc.
 - **Thesis:** Adaptive Multi-task Medical Image Restoration, Supervised by Prof.Yu-Mei Huang and rated as A⁺.

EXPERIENCE

- **Tencent Hunyuan** Shenzhen, China
Research Intern (Qingyun Project Up) Mar. 2026 – Present
 - Research on the omnimodal foundation model, focusing on omnimodal video understanding.
- **SenseTime Research** Shenzhen, China
Research Intern, General Perceptual Computing Group, Advisor: Dr. Yichao Wu Feb. 2025 – Feb. 2026
 - Research on streaming MLLMs, focusing on hallucination mitigation and consistent vision-text alignment.
- **Heriot-Watt University** Edinburgh, UK
Remote Research Intern, Advisor: Prof.Wei Pang and Dr.Yingfang Yuan March. 2024 – Present
 - Research on diffusion-based image generation and multi-agent systems for task-specialized visual synthesis.
- **KAUST** Saudi Arabia
Remote Research Intern, Advisor: Dr.Jun Chen and Prof.Mohamed Elhoseiny Dec. 2023 – Jan. 2026
 - Exploring multi-agent coordination for web-scale retrieval to augment the reasoning capability of MLLMs.
- **The Chinese University of Hong Kong, Shenzhen** Shenzhen, China
Research Assistant, Advisor: Prof.Ruimao Zhang Apr. 2024 – Nov. 2024
 - Developed MLLMs' alignment between visual and textual cues to enable human-centric multimodal generation.
- **iFLYTEK Co., Ltd.** June 2023 - Aug. 2023
Data Analysis Assistant Intern Lanzhou, China
 - Developed multimodal recommendation models from multi-source behavior-aware cues.

PAPERS AND PUBLICATIONS(CORE A* × 12)

C = Conference, J=Journal, T = Tech Report, R = Under Review

* means Corresponding Author, † means equal contribution

Multimodal Understanding, Agents & Efficient MLLMs

- [C.1] Zhongyu Yang, Junhao Song, Siyang Song, Wei Pang, Yingfang Yuan. **MERMAID: Multi-perspective Self-reflective Agents with Generative Augmentation for Emotion Recognition**. EMNLP Main 2025 (**Core A***).
- [C.2] Zhongyu Yang[†], Yingfang Yuan[†], Xuanming Jiang, Baoyi An, Wei Pang. **InEx: Hallucination Mitigation via Introspection and Cross-Modal Multi-Agent Collaboration**. AAAI 2026, (Acceptance Rate = 17.6%, **Core A***).
- [C.3] Zhongyu Yang, Wei Pang, Yingfang Yuan. **X^R: Cross-Modal Agents for Composed Image Retrieval**. WWW 2026 (Acceptance Rate = 20.1%, **Core A***).
- [C.4] Zhongyu Yang, Zuhao Yang, Wei Pang, Yingfang Yuan. **SVAgent: Storyline-guided Long Video Understanding via Cross-modal Multi-agent Collaboration**. CVPR 2026 (**Core A***).
- [C.5] Zhongyu Yang, Dannong Xu, Yonghan Zhang, Kefan Chen, Xinyi Wang, Yang Xu, Wei Pang, Yingfang Yuan. **Do Vision and Text Cues Exhibit Evidential Coupling? UFO: A Benchmark for Compositional Multimodal Reasoning in Unified Models**. ICML 2026 (**Core A***).
- [C.6] Dannong Xu[†], Zhongyu Yang[†], Jun Chen, Yingfang Yuan, Ming Hu, Lei Sun, Luc Van Gool, Danda Pani Paudel, Chun-Mei Feng. **MultiHaystack: Benchmarking Multimodal Retrieval and Reasoning over 40K Images, Videos, and Documents**. ECCV 2026 (**Core A***).

[C.7] Zhengwei Zou, Xuanming Jiang, Dingyu Nie, Zhengxing Fang, Qingyu Liu, Xueming Qian, Baoyi An, Guoshuai Zhao, **Zhongyu Yang***. **EmoRes: Toward User-Agnostic Psychological Support via Topic-Mining Emotional Agent.** *ACL 2026* (**Core A***).

[J.1] **Zhongyu Yang[†]**, Dannong Xu[†], Wei Pang, Yingfang Yuan. **Script: Graph-Structured and Query-Conditioned Semantic Token Pruning for Multimodal Large Language Models.** *Transactions on Machine Learning Research*.

[R.1] **Zhongyu Yang**, Jiale Tao, Ruitao Chen, Zuhao Yang, Yingfang Yuan, Xueliang Zhao, Auden, Kai Wang, Shuai Shao, Biao Wang, Steve Yves, Qinglin Lu. **OmniCapBench: A Deep-Structured Evaluation Framework for Fine-Grained Audio-Visual Captioning.** Under review.

Multimodal Generation & Creative Applications

[C.1] **Zhongyu Yang[†]**, Jun Chen[†], Dannong Xu, Junjie Fei, Xiaoqian Shen, Liangbing Zhao, Chun-Mei Feng, Mohamed Elhoseiny. **WikiAutoGen: Towards Multi-Modal Wikipedia-Style Article Generation.** *ICCV 2025* (**Core A***).

[C.2] **Zhongyu Yang**, Junhao Song, Zhang Luo, Zuhao Yang, Yang Xu, Jingfen Lan, Yonghan Zhang, Wei Pang, Siyang Song, Yingfang Yuan. **ReChar: Revitalizing Characters with Structure-Preserved and User-Specified Aesthetic Enhancements.** *Siggraph Asia 2025 Oral* (**Core A***).

[C.3] Xuanming Jiang, Baoyi An, Dingyu Nie, Haoyu Ren, Zhengwei Zou, Yizhe Yang, Jialie Shen, Zhiwen Jin, Xueming Qian, **Zhongyu Yang***, Guoshuai Zhao. **Bridge Your Fields: Non-Uniform Meteorological Field Reconstruction.** *CVPR 2026* (**Core A***).

[C.4] Junhao Song, Lluís Guasch, Xilin He, **Zhongyu Yang**, Yingfang Yuan, Weicheng Xie, Linlin Shen, Lu Liu, Wei Pang, Siyang Song. **CHAT: Conversational Human Audio-visual Talking Dialogue Generation.** *ECCV 2026* (**Core A***).

[T.1] Keming Wu, Zuhao Yang, Kaichen Zhang, Shizun Wang, Haowei Zhu, Sicong Leng, **Zhongyu Yang**, Qijie Wang, Sudong Wang, Ziting Wang, Zili Wang, Hui Zhang, Haonan Wang, Hang Zhou, Yifan Pu, Xingxuan Li, Fangneng Zhan, Bo Li, Lidong Bing, Yuxin Song, Ziwei Liu, Wenhui Chen, Jingdong Wang, Xinchao Wang, Xiaojuan Qi, Shijian Lu, Bin Wang. **Visual Generation in the New Era: An Evolution from Atomic Mapping to Agentic World Modeling.**

[T.2] Tencent Hunyuan Team (including **Zhongyu Yang**). **Script-a-Video: Deep Structured Audio-visual Captions via Factorized Streams and Relational Grounding.**

Economics, Math & Others

[J.1] Mengying Su, **Zhongyu Yang***, Shujaat Abbas, Yuriy Bilan, Agnieszka Majewska. **Toward Enhancing Environment Quality in OECD Countries: Role of Municipal Waste, Renewable Energy, Environment Innovation and Environmental Policy.** *Renewable energy* (**SCI Q1 Top, IF=9.0**).

[J.2] Zhichao Yu, Wenlan Xie, Junjie Guo, **Zhongyu Yang***. **Green Effect of Energy Transition Policy: A quasi-natural Experiment Based on New Energy Demonstration Cities.** *Finance Research Letters* (**SSCI Q1 Top, IF=10.4**).

[C.1] Yizhe Yang, Xuanming Jiang, Jisheng Dang, Aoying Wang, Baoyi An, Hao Wu, Guoshuai Zhao, Bimei Wang, Hong Peng, Bin Hu, **Zhongyu Yang***. **SCOPE: Streaming Covariance-Orthogonal Patching for Efficient LLM Safety Governance.** *KDD 2026* (**Core A***).

PATENTS

P=Patent, S=Software

[P.1] **Zhongyu Yang**. **A mathematics teaching system based on virtual reality.** (CN116312091A)

[S.1] **Zhongyu Yang**. **Green and Low-carbon Integrated Monitoring Software.**(2023SR1355487)

[S.2] **Zhongyu Yang**. **Fully automatic spatial sound field environment perception system.** (2024SR0538446)

HONORS AND AWARDS

• Excellence Graduation thesis, Lanzhou University (2025)	Top 5%
• Best Wiki Winner, International Directed Evolution Competition (IDEC) (2024)	Top 5%
• Meritorious Winner, ICM Mathematical Modeling Competition (2023)	Top 6%
• Honorable Mention, MCM (2023), focus on multimodal data modeling	Top 25%
• Best Hardware & Target Design, IDEC (2023), integrating ML with real-world application	Top 1%
• 1st Prize, National Data Analysis Competition (2022), applied to visual data regression	Top 3%
• Outstanding Student Scholarship, Lanzhou University (2022, 2024)	Top 15%

SKILLS AND SERVICES

- **Programming Languages:** Python, R, Stata, Latex
- **Languages:** Mandarin (Native), Cantonese (Native), English (Fluent)
- **Operation System:** Windows (advanced), Linux (advanced)
- **Reviewer Services:** CVPR, ICCV, ECCV, ICML, ICLR, Siggraph Asia, ARR(ACL, EMNLP, NAACL, TACL), AAAI, ICME; EMFT (Q1), ESPR (Q1), IJER (Q2)